1. Objective

Constraints help keep data correct and maintain table relationships.

2. Constraints and Purpose

Constraint	Example	Purpose
PRIMARY KEY	Passenger_ID	Uniquely identifies each row.
FOREIGN KEY	Booking.Passenger_ID	Prevents bookings for missing passengers.
UNIQUE	Passenger.Email	Prevents duplicate email values.
UNIQUE composite	Booking(Flight_ID, Seat_ID)	Prevents duplicate seat allocation on one flight.
CHECK	Aircraft.Capacity > 0	Prevents invalid capacity.
CHECK	Arrival_Time > Departure_Time	Ensures arrival is after departure.
NOT NULL	Passenger.Name	Requires important information.
DEFAULT	Booking_Status = Confirmed	Adds a standard value when omitted.

3. Test Constraints

```
-- Duplicate email: should fail UNIQUE\nINSERT INTO Passenger VALUES\n(99,'Test Person','aarav@example.
com','9000000000','P999999');\n\n-- Invalid capacity: should fail CHECK\nINSERT INTO Aircraft VALUES (99,
'Test Plane',0,'Maker');\n\n-- Missing passenger: should fail FOREIGN KEY\nINSERT INTO Booking\n(
Booking_ID,Passenger_ID,Flight_ID,Seat_ID,Booking_Date,Booking_Status)\nVALUES (5099,999,1001,103,'2026-
10-24','Confirmed');
```

Expected result: the three invalid inserts are rejected. Exact error messages depend on the MySQL version. CHECK constraints are enforced in MySQL 8.0.16 and later.

4. Check Relationships

```
SHOW CREATE TABLE Booking;\nSHOW CREATE TABLE Payment;\nSELECT * FROM Booking WHERE Passenger_ID NOT IN\
n (SELECT Passenger_ID FROM Passenger);
```

The last query should return no rows when all booking passenger references are valid.

5. Conclusion

Primary keys, foreign keys, UNIQUE, CHECK, NOT NULL and DEFAULT constraints prevent invalid or duplicate data.